Lecture 7

Tuesday, 31 January 2023 12:02 PM

$$\frac{d\underset{\sim}{x}}{dt} = \underset{\sim}{K}\underset{\sim}{x} \quad ; \quad \underset{\sim}{K} \equiv \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1n} \\ \vdots & & \ddots & \\ k_{n1} & & & k_{nn} \end{bmatrix} \quad ; \quad k_{ii} = -\sum_{j}^{n} k_{ji}$$

$\underset{\sim}{K}\underset{\sim}{x} = \underset{\sim}{0}$; $k_{ij} x_j^* = k_{ji} x_i^*$ → principle of microscopic reversibility

$$\sqrt{\underset{\sim}{D}} = \begin{bmatrix} \sqrt{x_1^*} & \cdots & 0 \\ & \sqrt{x_2^*} & \\ \vdots & & \ddots \\ 0 & & \sqrt{x_n^*} \end{bmatrix} \quad ; \quad \left(\sqrt{\underset{\sim}{D}}\right)^{-1} \underset{\sim}{K} \left(\sqrt{\underset{\sim}{D}}\right) = \underset{\sim}{K}$$

$\underset{\sim}{B}^{-1}\underset{\sim}{A}\underset{\sim}{B} = \underset{\sim}{C}$; A & C have the same

$\underset{\sim}{B}^{-1}\underset{\sim}{A}\underset{\sim}{B} - \lambda\underset{\sim}{B}^{-1}\underset{\sim}{B} = \underset{\sim}{C} - \lambda I$

$\underset{\sim}{B}^{-1}(\underset{\sim}{A} - \lambda\underset{\sim}{I})\underset{\sim}{B} = \underset{\sim}{C} - \lambda\underset{\sim}{I}$

$|(A - \lambda I)| = |C - \lambda I| = 0$

$\underset{\sim}{z}^{\dagger}\underset{\sim}{K}\underset{\sim}{z} \leq 0 \qquad z_j \underset{\sim}{D}^{-1} z_k = 0 \qquad j \neq k$

$$\underline{x} = \sum_{j=0}^{n-1} \alpha_j \underline{z}_j \quad ; \quad \underline{z}_0 = \underset{\sim}{x}^*$$

$$\left(\sqrt{D}\right)^{-1} K \left(\sqrt{D}\, z_j\right) = \tilde{\underset{\sim}{K}}\, \tilde{z}_j \quad ; \text{ given } z_j \underset{\sim}{D}^{-1} z_k = 0$$

We can find α_j : $\boxed{\alpha_j = \dfrac{z_j \underset{\sim}{D}^{-1} x}{z_j \underset{\sim}{D}^{-1} z_j}}$

$$\underset{\sim}{x}_0 = \sum_{k=0}^{n-1} \frac{(x_0 \underset{\sim}{D}^{-1} z_k)}{(z_k \underset{\sim}{D}^{-1} z_k)} z_k e^{\lambda_k t}$$

Parameter Estimation in Nonlinear Kinetics

Determining kinetic parameters : $r \rightarrow$ rate

$\xi \rightarrow$ extent

$k \rightarrow$ rate constants $k = [k_1\ k_2 \cdots k_p] \rightarrow p$ parameters

$\frac{d\xi}{dt} = r(\xi, k)$; $\xi(0) = 0$ → ODE representing the temporal evolution of ξ

$\{\hat{\xi}_i\}_{i=1}^{N}$: $\underset{k \in \mathbb{R}^p}{Min} = \sum_{i=1}^{N} |\xi_i(t,k) - \hat{\xi}_i|^2$

prediction vs. measured

Bellman's elegant quasi-linearization approach

$m^{th} \rightarrow (m+1)^{th}$

$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$; $n = p+1$; $x_1 = \xi$, $x_2 = k_1, \ldots, x_n = k_p$ (Redefining the variables)

$f_1(x) = r(\xi, k)$; $f_2(x) = 0$. ⇒ $\underline{f}$

→ $\frac{d\underline{x}}{dt} = \underline{f}(x)$ → New ODE

Jacobian (derivative w.r.t each variable)

Linearize this → $\frac{dx^{(m+1)}}{dt} = \underline{f}(x^{(m)}) + (\nabla \underline{f}(x^{(m)}))(x^{(m+1)} - x^{(m)})$; Let $J^{(m)} = \nabla \underline{f}(x^m)$

↳ particular soln: $\frac{dp}{dt} = \underline{f}(x^{(m)}) + J^{(m)}(p - x^{(m)})$; $p(0) = 0$

↳ Homogeneous soln: $\frac{dh^j}{dt} = J^{(m)} h^{(j)}$; $h^j(0) = e^j = [0 \ldots 1 \ 0 \ldots 0]^T$

$\therefore x^{(m+1)}(t) = p(t) + \sum_{j=1}^{n} x_j(0) h^j(t)$; p unknowns $\{x_j(0)\}$

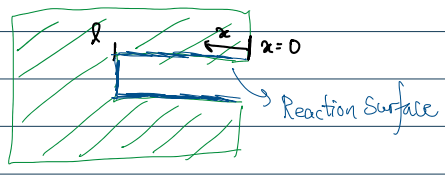
Minimize first component → extent of reaction

$\sum_{i=1}^{N} \left[p_1(t_i) - \sum_{j=1}^{n} x_j(0) h^j(t_i) - \hat{\xi}_i \right]^2$ → Iterate until satisfactory convergence.

Transport Effects in Catalytic Reaction Systems

Single Pore Analysis:

+ Steady state
+ First order
+ Isothermal



$$-DA\left.\frac{dc}{dx}\right|_x - \left(-DA\left.\frac{dc}{dx}\right|_{x+dx}\right) + \hat{K}_c P dx = 0$$

$\hat{K}_c P dx$ ↓ reaction

Diffusive flux

BC's: 1) @ $x=0$, $c=c_0$

2) @ $x=l$, $\frac{dc}{dx}=0$

$$\therefore D\frac{d^2c}{dx^2} + \frac{P}{A}\hat{K}_c = 0$$

$0 < x < l$

Whatever diffuses in the pore reacts: so $DA\left.\frac{dc}{dx}\right|_{x=0}$ → Total reaction rate in pore

Reaction rate in pore at uniform conc: $c_0 = Pl\hat{K}c_0$

Effectiveness factor Matrix, N_i: $NPl\hat{K}c_0 = \left[DA\frac{dc}{dx}\right]_{x=0}$

Non-dim: $\xi = \frac{x}{l}$; $\phi^2 = \frac{-Pl^2D^{-1}\hat{K}}{A}$

$$\therefore \frac{d^2c}{d\xi^2} = \phi^2 c$$

$c = [\cosh\phi(1-\xi)]k_1 + [\sinh\phi(1-\xi)]k_2$; k_1 & k_2 are constants